

Explainability

The property of a system whose outputs can be accounted for in humanly intelligible terms

GLOSSARY 06 - SHEET 4 / 8

PROFESSIONAL DEFINITION

Explainability is the property of an AI system whose behaviour and specific outputs can be accounted for in terms intelligible to a relevant human audience. It is distinguished from interpretability, which concerns whether the model's internal mechanism is directly inspectable, and it is audience-relative: the explanation adequate for a data scientist differs from that required by a decision subject or a regulator (Guidotti et al., 2018; Rudin, 2019).

WHO DEFINED IT ORIGINALLY

The research field was named and funded through DARPA's Explainable AI programme from 2016. Riccardo Guidotti and colleagues (2018) supplied the systematic taxonomy of methods in ACM Computing Surveys. Cynthia Rudin of Duke University (2019) argued in Nature Machine Intelligence that for high-stakes decisions inherently interpretable models should be used instead of post hoc explanations of opaque ones.

WHY THIS DEFINITION WAS DEVELOPED

Accurate models were being deployed in consequential decisions without any account of the basis for individual outcomes, which prevented contest, correction and audit. Explainability was needed to make the individual decision reviewable; Rudin's intervention added the argument that a post hoc explanation of an opaque model is an approximation of the model rather than the reason for the decision.

HOW IT IS USED

Specify the audience and the purpose before selecting a method, and prefer an inherently interpretable model where the decision is consequential, since a post hoc explanation may be faithful to the model's behaviour without being the actual reason for the outcome.

WHEN IT IS USED

At the design of any system whose outputs affect individuals, when a decision must be contested or appealed, in conformity assessment of a high-risk system, and in model validation.

BY WHOM IT IS USED

Data scientists selecting model families, model risk and validation functions, compliance teams evidencing transparency duties, and the decision subjects entitled to an account.

FIGURE 1 - EXPLANATION BY AUDIENCE AND SCOPE

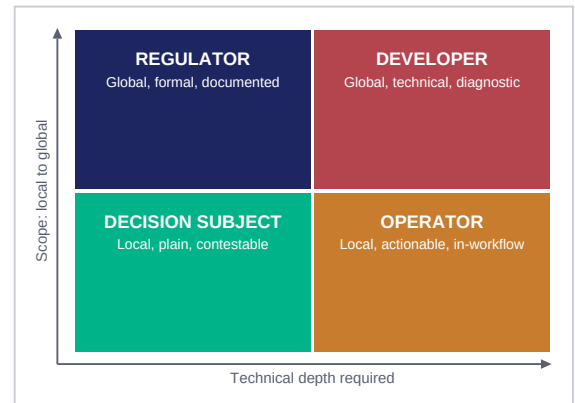


Figure 1. Explainability is audience-relative and no single artefact serves all four. A feature-importance plot satisfies the developer and is not an account of why this applicant was refused, which is what the decision subject is entitled to (after Guidotti et al., 2018).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor position accepted opacity as the price of accuracy and offered global feature-importance measures as a substitute for individual explanation. The field's contribution is to distinguish global from local explanation and model-agnostic from model-specific methods, so the right instrument can be matched to the question asked. Rudin's argument then challenges the accuracy-interpretability trade-off itself, holding that on structured data the trade-off is frequently negligible and the opacity therefore unnecessary.

RELATED DEFINITIONS IN THIS COURSE

[Algorithmic Bias - Unit 6](#)
[Accountability Gap - Unit 6](#)
[Human-in-the-Loop - Unit 2](#)
[AI Governance - Unit 6](#)
[Algorithm Aversion - Unit 2](#)

SOURCES (HARVARD REFERENCING STYLE)

- Guidotti, R., Monreale, A., Ruggieri, S., Turini, F., Giannotti, F. and Pedreschi, D. (2018) 'A survey of methods for explaining black box models', ACM Computing Surveys, 51(5), pp. 1-42.
- Gunning, D. and Aha, D.W. (2019) 'DARPA's explainable artificial intelligence program', AI Magazine, 40(2), pp. 44-58.
- Rudin, C. (2019) 'Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead', Nature Machine Intelligence, 1(5), pp. 206-215.